

EFGCL: Learning Dynamic Motion through Spotting-Inspired External Force Guided Curriculum Learning

Keita Yoneda¹, Kento Kawaharazuka^{1,2}, Kei Okada¹

Abstract— Learning dynamic whole-body motions for legged robots through reinforcement learning (RL) remains challenging due to the high risk of failure, which makes efficient exploration difficult and often leads to unstable learning.

In this paper, we propose External Force Guided Curriculum Learning (EFGCL), a guided RL approach based on the principle of *physical guidance*, in which external assistive forces are introduced during training. Inspired by spotting in artistic gymnastics, EFGCL enables agents to physically experience successful motion executions without relying on task-specific reward shaping or reference trajectories.

Experiments on a quadrupedal robot performing Jump, Backflip, and Lateral-Flip tasks demonstrate that EFGCL accelerates learning of the Jump task by approximately a factor of two and enables the acquisition of complex whole-body motions that conventional RL methods fail to learn. We further show that the learned policies can be deployed on a real robot, reproducing motions consistent with those observed in simulation.

These results indicate that physically guided exploration, which allows agents to experience success early in training, is an effective and general strategy for improving learning efficiency in dynamic whole-body motion tasks.

I. INTRODUCTION

To achieve high locomotion performance in unstructured environments, quadrupedal robots require learning methods that can stably acquire a wide range of diverse and complex motor skills. With recent advances in reinforcement learning (RL), numerous approaches have been proposed that enable robust learning of individual behaviors, such as locomotion over rough terrain [1]–[5].

For tasks such as rough-terrain locomotion and obstacle traversal, learning methods with high success rates have already been established. In contrast, learning dynamic motor skills involving high acceleration and high energy, as exemplified by sports motions [6], still requires substantial task-specific tuning. This difficulty arises because motions with a high risk of failure are inherently difficult to explore, and learning rarely progresses without explicit guidance.

To address the learning of such dynamic motions, Guided Reinforcement Learning (Guided-RL), which introduces assistance during the learning process, has been widely studied [7]. Representative approaches include imitation learning based on reference trajectories and reward shaping, which guides behavior through carefully designed reward functions.

¹ The authors are with the Department of Mechano-Informatics, Graduate School of Information Science and Technology, The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo, 113-8656, Japan. [yoneda, kawaharazuka, k-okada]@jsk.imi.i.u-tokyo.ac.jp

² The author is with the AI Center, Graduate School of Information Science and Technology, The University of Tokyo, Japan.

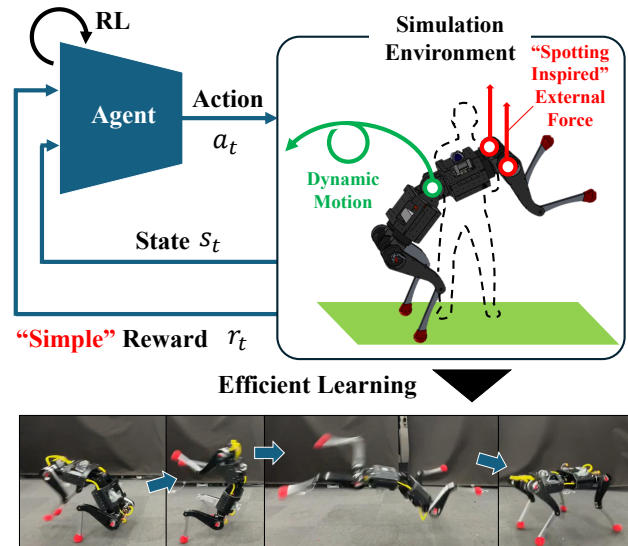


Fig. 1. Conceptual overview of External Force Guided Curriculum Learning (EFGCL). By applying external assistive forces in the early stages of learning, the agent experiences motion sequences with a higher probability of success. As learning progresses, the assistance is gradually reduced in a curriculum manner, ultimately enabling the agent to acquire a policy that achieves the target motion without assistance.

Imitation learning directly mimics reference trajectories that represent target motions and has been increasingly applied to dynamic tasks [8]–[10]. However, the performance of the learned motions strongly depends on the quality of the reference trajectories. Methods such as Opt-Mimic [11], which generate trajectories through optimization, can provide high-quality references, but they incur substantial costs in robot modeling and objective function design. Alternatively, approaches such as WASABI [12] utilize demonstrations obtained by physically guiding the robot, reducing data collection costs. However, the quality of such data depends heavily on the skill of the human operator, making it difficult to ensure stability. As a result, there exists an inherent trade-off between the quality of reference trajectories and the cost of generating them, and achieving high performance solely through imitation learning remains expensive.

Reward shaping aims to facilitate exploration by designing intermediate rewards that capture key elements of the desired behavior [13]–[15]. However, determining which aspects of a motion should be defined as intermediate rewards is highly task-dependent and non-trivial. Moreover, intermediate rewards may introduce designer bias, which can exclude potentially optimal motion sequences and degrade learning performance [16]. For these reasons, [16] recommends using sparse reward functions and promoting learning through

EFGCL：通过保护启发的外力引导课程学习实现动态运动学习

Keita Yoneda¹, Kento Kawaharazuka^{1,2}, Kei Okada¹

摘要— 通过强化学习（RL）为腿式机器人学习动态全身运动仍然具有挑战性，因为存在高失败风险，这使得高效探索变得困难，并常常导致学习过程不稳定。

在本文中，我们提出了外力引导课程学习（EFGCL），这是一种基于物理引导原理的引导式强化学习方法，在训练过程中引入外部辅助力。受竞技体操中保护的启发，EFGCL使智能体能够实际体验成功的运动执行，而无需依赖特定任务的奖励塑形或参考轨迹。

在四足机器人执行跳跃、后空翻和侧空翻任务的实验中，EFGCL将跳跃任务的学习速度提升了约两倍，并能够习得传统强化学习方法无法学会的复杂全身运动。我们进一步展示了所学策略可部署到真实机器人上，复现出与仿真中观察到的运动一致的行为。

这些结果表明，物理引导探索能够让智能体在训练早期体验成功，是提高动态全身运动任务学习效率的一种有效且通用的策略。

I. 引言

为了在非结构化环境中实现高水平的运动性能，四足机器人需要能够稳定获取各种多样且复杂的运动技能的学习方法。随着强化学习（RL）的最新进展，已经提出了许多方法，使得能够稳健地学习个体行为，例如在崎岖地形上的运动 [1]–[5]。

对于崎岖地形运动和障碍穿越等任务，具有高成功率的学习方法已经建立。相比之下，学习涉及高加速度和高能量的动态运动技能，如体育动作 [6]，仍然需要大量的任务特定调优。这一困难源于高失败风险的运动本质上难以探索，且在没有明确指导的情况下，学习很少取得进展。

为了解决此类动态运动的学习问题，引导强化学习（Guided-RL）引入了—在学习过程中提供阻力，已被广泛研究[7]。代表性方法包括基于参考轨迹的模仿学习和奖励塑形，后者通过精心设计的奖励函数来引导行为。

¹ 作者隶属于东京大学信息科学与技术研究生院机械信息学系，地址：日本东京都文京区本乡7-3-1，邮编113-8656。[yoneda, kawaharazuka, k-okada]@jsk.imi.i.u-tokyo.ac.jp² 作者隶属于东京大学信息科学与技术研究生院人工智能中心，日本。

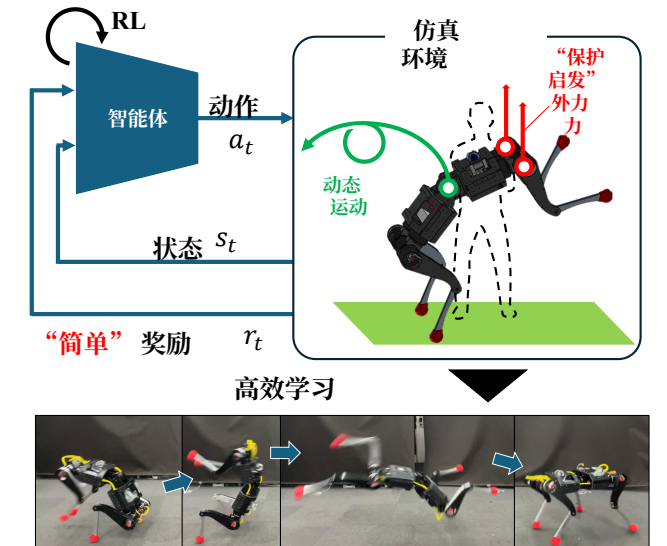


图1. 外力引导课程学习（EFGCL）的概念总览。通过在学习的早期阶段施加外部辅助力，智能体能够以更高的成功率体验运动序列。随着学习的推进，辅助力按照课程学习的方式逐渐减少，最终使智能体获得无需辅助即可实现目标运动的策略。

模仿学习直接模仿代表目标运动的参考轨迹，并已越来越多地应用于动态任务 [8]–[10]。然而，学习到的运动性能在很大程度上取决于参考轨迹的质量。诸如 Opt-Mimic [11]等通过优化生成轨迹的方法可以提供高质量的参考，但它们在机器人建模和目标函数设计方面会产生大量成本。另外，像WASABI [12]这样的方法利用通过物理引导机器人获得的演示，降低了数据收集成本。然而，此类数据的质量在很大程度上取决于操作人员的技能水平，难以确保稳定性。因此，参考轨迹的质量与其生成成本之间存在固有的权衡，仅通过模仿学习实现高性能仍然代价高昂。

奖励塑形旨在通过设计中间奖励来促进探索，这些中间奖励捕捉了期望行为的关键要素 [13]–[15]。然而，确定运动的哪些方面应被定义为中间奖励高度依赖于任务且并非易事。此外，中间奖励可能引入设计者偏差，从而排除潜在的最优运动序列并降低学习性能 [16]。因此，[16] 建议使用稀疏奖励函数，并通过奖励之外的设计选择来促进学习

design choices outside the reward itself. Nevertheless, a general framework for efficiently exploring dynamic motions with a high risk of failure has yet to be established.

Insightful inspiration can be drawn from artistic gymnastics, where dynamic motions are the primary objective. In gymnastics training, a technique known as *spotting* is commonly used, in which a coach physically supports the athlete while practicing a skill [17]. This approach assists the exploration process through physical guidance and differs fundamentally from conventional methods that guide behavior solely through reward design.

Motivated by this idea, we propose *External Force Guided Curriculum Learning* (EFGCL), which applies this principle to reinforcement learning for robots (Fig. 1). The main contributions of this work are summarized as follows:

- We introduce a new learning paradigm for dynamic motor skill acquisition that employs *physical guidance via external forces*, rather than guidance through reward design.
- We propose External Force Guided Curriculum Learning (EFGCL), which gradually decays external assistive forces and demonstrate that physical guidance significantly improves exploration efficiency.
- Through learning experiments on a real quadrupedal robot, we demonstrate that the proposed approach is effective and transferable to real-world environments.

II. BACKGROUND

A. Proximal Policy Optimization (PPO)

In reinforcement learning for legged robots, Proximal Policy Optimization (PPO) [18] is commonly used due to its training stability and ease of implementation. PPO constrains the magnitude of policy updates by optimizing a clipped surrogate objective function based on the likelihood ratio between the current and previous policies,

$$r_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}, \quad a_t \sim \pi_{\theta_{\text{old}}}(\cdot | s_t),$$

thereby enabling stable learning while limiting excessive policy updates. This update control is particularly effective for legged robots with high-dimensional action spaces and unstable dynamics, as it prevents training collapse caused by abrupt policy changes.

PPO also employs Generalized Advantage Estimation (GAE) [19] to estimate the advantage A_t . GAE computes the advantage as a weighted sum of temporal-difference (TD) errors, $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$, where $\gamma \in [0, 1]$ is the discount factor. This formulation allows a trade-off between variance and bias. Based on the estimated A_t , PPO updates the policy in the following gradient direction:

$$\nabla_\theta J(\theta) \propto \mathbb{E}_{s_t, a_t \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a_t | s_t) A_t].$$

However, in environments with a high risk of failure, unsuccessful trajectories tend to dominate, resulting in small estimated state values $V(s_t)$ for many states. Consequently, both the TD error δ_t and the advantage A_t approach zero, providing little useful information for gradient-based updates.

Algorithm 1 External Force Guided Curriculum Learning

- 1: **Input:** initial policy π_{init} , initial critic V_{init} , assist force F_{assist} , decay step size ε , success threshold ζ
- 2: **Output:** final policy π_N
- 3: Initialize:

$$F_0 \leftarrow F_{\text{assist}}, \quad \pi_0 \leftarrow \pi_{\text{init}}, \quad V_0 \leftarrow V_{\text{init}}$$

- 4: Set decay rate:

$$\alpha \leftarrow 1.0$$

- 5: **for** $i = 0, \dots, N-1$ **do**
- 6: $\text{success_rate} \leftarrow 0$
- 7: **while** $\text{success_rate} < \zeta$ **do**
- 8: Train π_i and V_i using PPO under assist force F_i
- 9: Update success_rate
- 10: **end while**
- 11: Decay assist force:

$$\alpha \leftarrow \max(0, 1 - \varepsilon \times i), \quad F_{i+1} \leftarrow \alpha \times F_{\text{assist}}$$

- 12: Carry over the learned policy and value function:

$$\pi_{i+1} \leftarrow \pi_i, \quad V_{i+1} \leftarrow V_i$$

- 13: **end for**
- 14: **return** π_N

Thus, while PPO is a stable optimization method, learning can stagnate severely in environments where successful experiences are rarely obtained.

B. Curriculum Learning

Curriculum learning has been widely adopted as a representative approach to address the aforementioned issue.

In curriculum learning, training begins in specialized environments with low exploration risk, and the task difficulty or risk level is gradually increased. This framework can be interpreted as sequential learning over multiple Markov Decision Processes (MDPs) $\mathcal{M}_i$ with different risk levels. If the changes in dynamics and reward structures between successive stages are sufficiently small, the optimal policy π_i^* for stage $\mathcal{M}_i$ and the optimal policy π_{i+1}^* for the subsequent stage $\mathcal{M}_{i+1}$ are expected to be similar. That is,

$$\frac{\pi_{i+1}^*(a_t | s_t)}{\pi_i^*(a_t | s_t)} \approx 1, \quad a_t \sim \pi_i^*(\cdot | s_t),$$

can be artificially constructed. This property is consistent with the assumption underlying PPO that smaller policy update steps are preferable. Therefore, curriculum learning provides a theoretically well-aligned framework that supports stable, incremental learning with PPO.

C. Stabilizing Learning via Accelerated Value Function Estimation

In reinforcement learning with sparse rewards, learning efficiency strongly depends on how quickly the value function can correctly evaluate action sequences that yield high rewards. This is particularly critical for methods such as

然而, 一个用于高效探索具有高失败风险的动态运动的通用框架尚未建立。

从艺术体操中可以汲取深刻的灵感, 在艺术体操中, 动态运动是主要目标。在体操训练中, 常用一种称为保护的技术, 即教练在运动员练习技能时提供身体支撑 [17]。这种方法通过物理引导辅助探索过程, 与仅通过奖励设计来引导行为的传统方法有着根本的不同。

受此启发, 我们提出了外力引导课程学习 (EFGCL), 将该原理应用于机器人强化学习 (图1)。本工作的主要贡献总结如下:

- 我们引入了一种新的动态运动技能习得学习范式, 该范式采用通过外力的物理引导, 而非通过奖励设计的引导。
- 我们提出了外力引导课程学习 (EFGCL), 该方法逐步衰减外部辅助力, 并证明物理引导显著提高了探索效率。
- 通过在真实四足机器人上的学习实验, 我们证明了所提出的方法是有效的, 并且可迁移到真实环境。

II. 背景

A. 近端策略优化 (PPO)

在腿式机器人的强化学习中, 近端策略优化 (PPO) [18] 因其训练稳定性和易于实现而被广泛使用。PPO通过基于当前策略与先前策略之间的似然比优化裁剪替代目标函数, 来约束策略更新的大小,

$$r_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}, \quad a_t \sim \pi_{\theta_{\text{old}}}(\cdot | s_t),$$

从而在限制过度策略更新的同时实现稳定学习。这种更新控制对于具有高维动作空间和不稳定动力学的腿式机器人尤为有效, 因为它可以防止因策略突变而导致的训练崩溃。

PPO还采用广义优势估计 (GAE) [19]来估计优势 A_t 。GAE将优势计算为时间差分 (TD) 误差的加权和, $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$, 其中 $\gamma \in [0, 1]$ 是折扣因子。这种公式允许在方差和偏差之间进行权衡。基于估计的 A_t , PPO沿以下梯度方向更新策略:

$$\nabla_\theta J(\theta) \propto \mathbb{E}_{s_t, a_t \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a_t | s_t) A_t].$$

然而, 在高失败风险的环境中, 不成功的轨迹往往占主导地位, 导致许多状态的估计状态值 $V(s_t)$ 较小。因此, TD误差 δ_t 和优势 A_t 都趋近于零, 为基于梯度的更新提供的有用信息很少。

算法1 外力引导课程学习

- 1: **输入:** 初始策略 π_{init} , 初始评论家 V_{init} , 辅助力 F_{assist} , 衰减步长 ε , 成功阈值 ζ
- 2: **输出:** 最终策略 π_N
- 3: 初始化:

$$F_0 \leftarrow F_{\text{assist}}, \quad \pi_0 \leftarrow \pi_{\text{init}}, \quad V_0 \leftarrow V_{\text{init}}$$

- 4: 设置衰减率:

$$\alpha \leftarrow 1.0$$

- 5: **对于** $i = 0, \dots, N-1$ **执行**
- 6: 成功率 $\leftarrow 0$
- 7: **当** 成功率 $< \zeta$ **时执行**
- 8: 在辅助力 F_i 下, 使用 PPO 训练 π_i 和 V_i
- 9: 更新成功率
- 10: **结束循环**
- 11: 衰减辅助力:

$$\alpha \leftarrow \max(0, 1 - \varepsilon \times i), \quad F_{i+1} \leftarrow \alpha \times F_{\text{assist}}$$

- 12: 继承学习到的策略和价值函数:

$$\pi_{i+1} \leftarrow \pi_i, \quad V_{i+1} \leftarrow V_i$$

- 13: **结束循环**
- 14: **返回** π_N

因此, 虽然PPO是一种稳定优化方法, 但在难以获得成功经验的环境中, 学习可能会严重停滞。

B. 课程学习

课程学习已被广泛采用, 作为解决上述问题的一种代表性方法。

在课程学习中, 训练始于探索风险较低的专业化环境, 任务难度或风险水平逐步提高。该框架可被理解为在多个具有不同风险水平的马尔可夫决策过程 (MDP) $\mathcal{M}_i$ 上的序列学习。如果连续阶段之间动态变化和奖励结构的差异足够小, 则阶段 $\mathcal{M}_i$ 的最优策略 π_i^* 与后续阶段 $\mathcal{M}_{i+1}$ 的最优策略 π_{i+1}^* 预计会相似。即,

$$\frac{\pi_{i+1}^*(a_t | s_t)}{\pi_i^*(a_t | s_t)} \approx 1, \quad a_t \sim \pi_i^*(\cdot | s_t),$$

可以被人为构建。这一性质与PPO所依据的假设一致, 即较小的策略更新步长更为可取。因此, 课程学习提供了一个理论上高度契合的框架, 支持使用PPO进行稳定、渐进的学习。

C. 通过加速价值函数估计稳定学习

在具有稀疏奖励的强化学习中, 学习效率在很大程度上取决于价值函数能够多快地正确评估产生高奖励的动作序列。这对于诸如

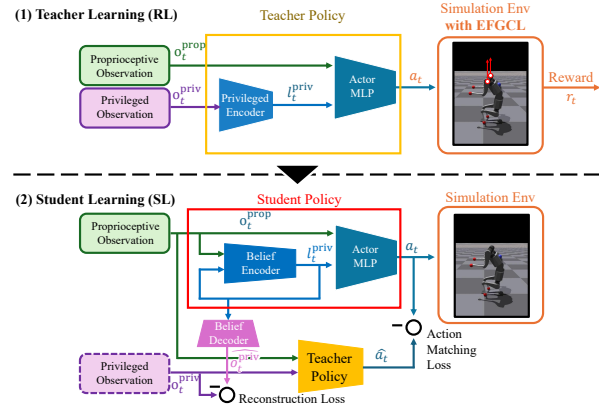


Fig. 2. Network architecture used for Teacher–Student learning. (a) The Teacher Policy takes the full state, including privileged observations, as input and is trained via reinforcement learning. (b) The Student Policy takes only onboard sensor information as input and is trained using the Teacher Policy as a supervisor.

PPO, which rely on value functions for advantage estimation, where the initial accuracy of value estimation significantly influences the direction of policy updates.

In general, once a sufficient number of high-reward trajectories are observed, the state values $V(s)$ corresponding to the associated state sequence $\xi = \{s_t^{\text{high}}\}_{t=0}^N$ are estimated to be large. When such high-value trajectories exist, actions that deviate from them yield large negative TD errors and advantages. As a result, the policy is updated in a direction that discourages deviation from high-reward trajectories, making effective motion sequences more likely to be preserved once acquired.

Therefore, whether the value function can assign high values to near-optimal behaviors from the early stages of learning is a crucial factor that determines the overall stability and efficiency of training. To achieve this, it is effective to expose the agent to a large number of high-reward trajectories during the early phase of learning.

III. METHOD

A. Overview of External Force Guided Curriculum Learning (EFGCL)

The proposed External Force Guided Curriculum Learning (EFGCL) provides a framework for achieving stable learning in dynamic tasks while maintaining sparse reward functions. Instead of modifying rewards as in imitation learning or Reward Shaping, EFGCL stabilizes learning by curriculum-wise modifying the Markov Decision Process (MDP) itself in which learning is performed.

The overall procedure of the algorithm is summarized in Alg. 1. EFGCL designs external assistive forces that facilitate task execution during the early stage of learning (line 1), and gradually decays this assistance as training progresses (lines 7–9). Through this curriculum process, the agent experiences trajectories with a high probability of success under assistance, while gradually transitioning toward autonomous motion generation. Eventually, the agent acquires a policy π_N that achieves the target motion without assistance (line 14).

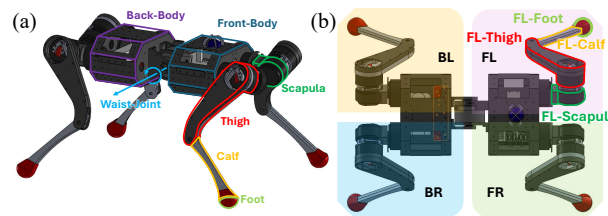


Fig. 3. Overview and kinematic structure of the quadrupedal robot KLEIYN. (a) Overall robot structure, (b) link and joint definitions for each leg.

B. Design of External Assistance

EFGCL first designs external assistive forces that help reproduce the target motions. As discussed in Sec. II-C, exposing the agent to a large number of high-reward trajectories in the early stage of learning induces a tendency to preserve such trajectories. The external assistance in EFGCL is introduced to artificially increase the density of these successful trajectories.

In EFGCL, external assistance is defined as a pattern consisting of three elements: the points of application $P = \{\mathbf{p}_i\}$, the corresponding force vectors $F = \{\mathbf{f}_i\}$, and the timing of application $T = \{(t_i^{\text{start}}, t_i^{\text{end}})\}$. Since the purpose of the assistance is not to teach an optimal trajectory but rather to guide the agent toward high-reward states, it is sufficient for the robot to approximately achieve the target motion and obtain high rewards. Therefore, in this study, the assistive force $F_{\text{assist}}(P, F, T)$ is heuristically designed for each task. The permissible range of such assistive forces is investigated in detail in Sec. V-B.

C. Success-Rate Based Adaptive Curriculum Scheduling

As discussed in Sec. II-B, curriculum learning benefits from small difficulty gaps between adjacent MDPs to ensure stable policy transitions. EFGCL adopts a success-rate-based adaptive curriculum to prevent excessive difficulty changes caused by curriculum updates.

Specifically, as shown in lines 6–12 of Alg. 1, PPO training is repeated at each stage i until the success rate exceeds a threshold ζ . The assistive force is then updated as

$$F_i = \alpha_i \times F_{\text{assist}}, \quad \alpha_i = \max(0, 1 - \varepsilon \times i).$$

where α_i denotes the assistance scaling factor at stage i , and ε is the decay step size controlling the rate of assistance reduction.

This mechanism enables automatic adjustment of the assistance decay step based on training progress, while preserving the relationship $\pi_{i+1}(a | s) / \pi_i(a | s) \approx 1$.

D. Time-Encoding for Observations

Since the assistive force $F_{\text{assist}}(P, F, T)$ in EFGCL is applied in a time-dependent manner, it is important for the agent to infer the timing of assistance as an internal state.

However, directly using the elapsed time t as an input leads to monotonically increasing values, which may cause scale mismatch in neural networks.

To address this issue, we introduce an additional observation that aligns with the activation interval of the assistive

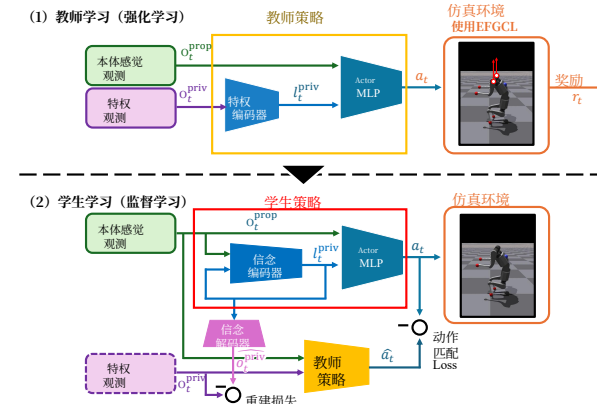


图2. 用于师生学习的网络架构。(a) 教师策略以全状态（包括特权观测）作为输入，并通过强化学习进行训练。(b) 学生策略仅以机载传感器信息作为输入，并以教师策略作为监督者进行训练。

PPO等方法尤为关键，这些方法依赖价值函数进行优势估计，其中价值估计的初始准确性显著影响策略更新的方向。

一般来说，一旦观察到足够数量的高奖励轨迹，与相关状态序列 $\xi = \{s_t^{\text{high}}\}_{t=0}^N$ 对应的状态值 $V(s)$ 就会被估计为较大值。当存在此类高价值轨迹时，偏离这些轨迹的动作会产生较大的负TD误差和优势值。因此，策略会朝着抑制偏离高奖励轨迹的方向更新，从而使有效的动作序列一旦习得就更有可能被保留。

因此，价值函数能否在学习早期阶段就为近最优行为赋予高价值，是决定整体训练稳定性和效率的关键因素。为了实现这一目标，在学习的早期阶段让智能体接触大量高奖励轨迹是有效的做法。

III. 方法

A. 外力引导课程学习（EFGCL）概述

所提出的外力引导课程学习（EFGCL）提供了一个框架，用于在保持稀疏奖励函数的同时，在动态任务中实现稳定学习。与模仿学习或奖励塑形中修改奖励的方式不同，EFGCL通过按课程方式修改进行学习的马尔可夫决策过程（MDP）本身来稳定学习。

算法的整体流程总结于算法1中。EFGCL在学习早期阶段设计外部辅助力以促进任务执行（第1行），并随着训练进度逐渐衰减该辅助（第7–9行）。通过这一课程过程，智能体在辅助下经历具有高成功概率的轨迹，同时逐步过渡到自主运动生成。最终，智能体获得策略 π_N ，无需辅助即可实现目标运动（第14行）。

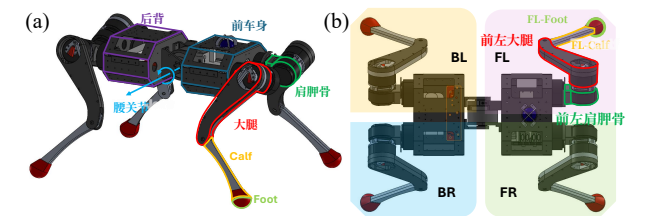


图3. 四足机器人KLEIYN的概述和运动学结构。(a) 整体机器人结构，(b) 每条腿的连杆和关节定义。

B. 外部辅助的设计

EFGCL首先设计有助于复现目标运动的外部辅助力。如第II-C节所述，在学习早期阶段让智能体接触大量高奖励轨迹，会诱导其倾向于保留此类轨迹。EFGCL中的外部辅助旨在人为增加这些成功轨迹的密度。

在EFGCL中，外部辅助被定义为由三个要素组成的模式：作用点 $P = \{\mathbf{p}_i\}$ 、相应的力向量 $F = \{\mathbf{f}_i\}$ 以及施加时机 $T = \{(t_i^{\text{start}}, t_i^{\text{end}})\}$ 。由于辅助的目的不是教授最优轨迹，而是引导智能体朝向高奖励状态，因此机器人只需大致实现目标运动并获得高奖励即可。因此，在本研究中，辅助力 $F_{\text{assist}}(P, F, T)$ 是针对每个任务启发式设计的。此类辅助力的允许范围将在第V-B节中详细研究。

C. 基于成功率的自适应课程调度

如第II-B节所述，课程学习受益于相邻MDP之间较小的难度差距，以确保策略转换的稳定性。EFGCL采用基于成功率的自适应课程，以防止课程更新导致的难度变化过大。

具体而言，如算法1第6–12行所示，在每个阶段重复进行PPO训练，直到成功率超过阈值 ζ 。随后，辅助力更新为

$$F_i = \alpha_i \times F_{\text{assist}}, \quad \alpha_i = \max(0, 1 - \varepsilon \times i).$$

其中 α_i 表示阶段 i 的辅助缩放因子， ε 是控制辅助减少速率的衰减步长。

该机制能够根据训练进度自动调整辅助衰减步长，同时保持关系 $\pi_{i+1}(a | s) / \pi_i(a | s) \approx 1$ 。

D. 观测的时间编码

由于EFGCL中的辅助力 $F_{\text{assist}}(P, F, T)$ 以时间依赖方式施加，智能体将辅助时机推断为内部状态至关重要。

然而，直接将经过时间 t 作为输入会导致单调递增的值，这可能引起神经网络中的尺度不匹配。

为了解决这个问题，我们引入了一个与辅助力的激活区间相一致的额外观测

force F_i . Specifically, we define a monotonically increasing function bounded within $[0, 1]$ as

$$\tau(t, \lambda) = \frac{\tilde{t}^3}{1 + \tilde{t}^3}, \quad \tilde{t} = \frac{t}{\lambda}.$$

Here, λ is a temporal scaling parameter, which is set to the force activation start time $\lambda = t^{\text{start}}$ in this study.

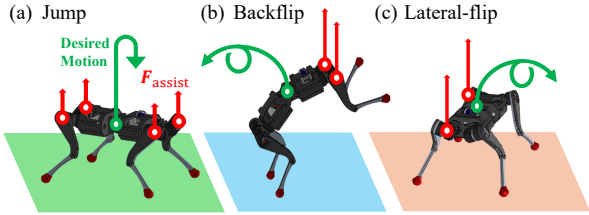


Fig. 4. Designed assistive force patterns for each task. (a) Jump, (b) Backflip, (c) Lateral-flip. The assistive forces are applied vertically to the scapula links, guiding motions that facilitate successful task execution during the early stages of learning.

IV. LEARNING SETUP

A. Robot Platform

The real-world experiments are conducted using the quadrupedal robot KLEIYN [6]. Its appearance and link definitions are shown in Fig. 3. KLEIYN has a total mass of 18 kg and a height of 600 mm, with three degrees of freedom (DoF) per leg and one DoF in the torso. The robot is equipped with an IMU and joint encoders. The leg motors are quasi-direct-drive actuators with a maximum torque of 24.8 Nm, while the torso motor has a maximum torque of 48 Nm. Isaac Gym [20] is used as the simulator for training.

B. Task Definition and Reward Function

To evaluate the effectiveness of EFGCL, we define three dynamic whole-body motion tasks: (1) Jump, (2) Backflip, and (3) Lateral-flip.

To avoid arbitrary performance gains due to task-specific reward engineering, all tasks share exactly the same reward structure, weights, and functional forms. Only the target variables differ between tasks: the maximum height for Jump, and the rotation angle for Backflip and Lateral-flip. These target variables are simply scaled according to their physical units, and no task-specific reward tuning or intermediate motion-guiding rewards are introduced.

The reward for each task is defined by the following common structure:

$$r_t = \rho_t^{\text{task}} + \rho_t^{\text{task}} \cdot \rho_t^{\text{stand}} + \lambda_{\omega} r_t^{\text{ang}} + r_t^{\text{common}}, \quad (1)$$

where $\rho_t^{\text{task}} \in [0, 1]$ represents task progress, ρ_t^{stand} encourages stable posture after landing, r_t^{ang} is a regularization term that suppresses rotation about non-target axes, and r_t^{common} enforces physical constraints shared across all tasks.

The only task-specific difference lies in the definition of the target quantity in ρ_t^{task} . For Jump, the target is the maximum achieved height, while for Backflip and Lateral-flip, the targets are the rotation angles around the pitch and roll axes, respectively. Detailed definitions are summarized in Appendix I.

C. Observations

The observation consists of two components, $\mathbf{o}_t^{\text{prop}}$ and $\mathbf{o}_t^{\text{priv}}$. $\mathbf{o}_t^{\text{prop}}$ represents proprioceptive observations and includes joint positions $\mathbf{q}_t \in \mathbb{R}^{13}$, joint velocities $\dot{\mathbf{q}}_t \in \mathbb{R}^{13}$, the gravity vector in the root frame $\tilde{\mathbf{g}}_t \in \mathbb{R}^3$, root angular velocity $\omega_t \in \mathbb{R}^3$, the command input $c \in \mathbb{R}$, and the time encoding $\tau(t) \in \mathbb{R}$. The command input corresponds to the target jump height h^{target} for Jump, and the target rotation angle θ^{target} for Backflip and Lateral-flip.

$\mathbf{o}_t^{\text{priv}}$ denotes privileged observations used exclusively by the Teacher Policy and consists of the following task-specific information:

- Jump: root height $h_t \in \mathbb{R}$ and maximum height since episode start $h_t^{\text{max}} \in \mathbb{R}$
- Backflip: root height h_t and root pitch angle $\theta_t^{\text{pitch}} \in \mathbb{R}$
- Lateral-flip: root height h_t and root roll angle $\theta_t^{\text{roll}} \in \mathbb{R}$

D. Assist Force Design

Assistive force patterns are designed for each task. The application timing is shared across all tasks as $T = \{(1.0\text{s}, 1.1\text{s})\}$, while the application points P and force vectors F are task-specific. The assistive force patterns are illustrated in Fig. 4.

a) Jump:

$$\begin{aligned} P^{\text{jump}} &= \{\mathbf{p}_{\text{FL}}^{\text{Scapula}}, \mathbf{p}_{\text{FR}}^{\text{Scapula}}, \mathbf{p}_{\text{BL}}^{\text{Scapula}}, \mathbf{p}_{\text{BR}}^{\text{Scapula}}\}, \\ F^{\text{jump}} &= \{(0, 0, f_{\text{jump}}(h^{\text{target}})/4)\} \end{aligned}$$

Here, $f_{\text{jump}}(h^{\text{target}})$ is the assistive force magnitude determined by the target height h^{target} . This value is derived by modeling the Jump motion as simple projectile motion and computing the average assistive force required to generate the initial velocity needed to reach the target height.

b) Backflip:

$$\begin{aligned} P^{\text{backflip}} &= \{\mathbf{p}_{\text{FL}}^{\text{Scapula}}, \mathbf{p}_{\text{FR}}^{\text{Scapula}}\}, \\ F^{\text{backflip}} &= \{(0, 0, 175\text{N})\} \end{aligned}$$

c) Lateral-flip:

$$\begin{aligned} P^{\text{lateral}} &= \{\mathbf{p}_{\text{FR}}^{\text{Scapula}}, \mathbf{p}_{\text{BR}}^{\text{Scapula}}\}, \\ F^{\text{lateral}} &= \{(0, 0, 300\text{N})\} \end{aligned}$$

E. Adaptive Curriculum Design

A success-rate-based curriculum scheduling strategy is employed to decay the assistive forces. The success rate in Alg. 1 is computed using the following criteria.

a) Jump:

$$|h_t^{\text{max}} - h^{\text{target}}| < 0.1 \quad \wedge \quad |h_t| < 0.1$$

b) Backflip, Lateral-flip:

$$|\theta_t - 2\pi| < 0.3 \quad \wedge \quad |h_t| < 0.1$$

The success rate threshold is set to $\zeta = 0.6$, and the decay step size is set to $\varepsilon = 0.01$ for all tasks.

F_{i0} . 具体来说，我们定义了一个在 $[0, 1]$ 内有界的单调递增函数，即

$$\tau(t, \lambda) = \frac{\tilde{t}^3}{1 + \tilde{t}^3}, \quad \tilde{t} = \frac{t}{\lambda}.$$

此处， λ 是时间缩放参数，在本研究中将其设置为力激活起始时间 $\lambda = t^{\text{start}}$ 。

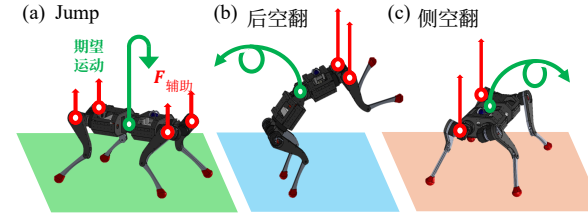


图4. 为每个任务设计的辅助力模式。(a) 跳跃, (b) 后空翻, (c) 侧空翻。辅助力垂直施加于肩胛骨连杆，引导运动以促进学习早期阶段的任务执行。

IV. 学习设置

A. 机器人平台

真实世界实验使用四足机器人KLEIYN [6]进行。其外观和连杆定义如图3所示。KLEIYN总质量为18kg，高度为600mm，每条腿具有三个自由度（DoF），躯干具有一个自由度。机器人配备了惯性测量单元和关节编码器。腿部电机采用准直驱执行器，最大扭矩为24.8Nm，而躯干电机的最大扭矩为48Nm。Isaac Gym [20] 用作训练模拟器。

B. 任务定义与奖励函数

为了评估EFGCL的有效性，我们定义了三个动态全身运动任务：（1）跳跃，（2）后空翻，以及（3）侧空翻。

为了避免因任务特定奖励工程而导致的任意性能提升，所有任务共享完全相同的奖励结构、权重和函数形式。只有目标变量在不同任务间有所不同：跳跃的最大高度，以及后空翻和侧空翻的旋转角度。这些目标变量仅根据其物理单位进行缩放，不引入任何任务特定奖励调优或中间运动引导奖励。

每个任务的奖励由以下通用结构定义：

$$r_t = \rho_t^{\text{task}} + \rho_t^{\text{task}} \cdot \rho_t^{\text{stand}} + \lambda_{\omega} r_t^{\text{ang}} + r_t^{\text{common}}, \quad (1)$$

其中 $\rho_t^{\text{task}} \in [0, 1]$ 表示任务进度， ρ_t^{stand} 鼓励落地后的稳定姿态， r_t^{ang} 是抑制非目标轴旋转的正则化项，而 r_t^{common} 强制执行所有任务共享的物理约束。

唯一的任务特定差异在于 ρ_t^{task} 中目标量的定义。对于跳跃，目标是达到的最大高度，而对于后空翻和侧空翻，目标分别是围绕俯仰轴和横滚轴的旋转角度。详细定义总结在附录I中。

C. 观测

观测由两个部分组成， $\mathbf{o}_t^{\text{prop}}$ 和 $\mathbf{o}_t^{\text{priv}}$ 。

$\mathbf{o}_t^{\text{prop}}$ 表示本体感觉观测，包括关节位置 $\mathbf{q}_t \in \mathbb{R}^{13}$ 、关节速度 $\dot{\mathbf{q}}_t \in \mathbb{R}^{13}$ 、根坐标系中的重力向量 $\tilde{\mathbf{g}}_t \in \mathbb{R}^3$ 、根角速度 $\omega_t \in \mathbb{R}^3$ 、命令输入 $c \in \mathbb{R}$ 以及时间编码 $\tau(t) \in \mathbb{R}$ 。命令输入对应于跳跃的目标跳跃高度 h^{target} ，以及后空翻和侧空翻的目标旋转角度 θ^{target} 。

$\mathbf{o}_t^{\text{priv}}$ 表示仅由教师策略使用的特权观测，包含以下任务特定信息：

- 跳跃：根高度 $h_t \in \mathbb{R}$ 以及回合开始以来的最大高度 $h_{\text{max}t} \in \mathbb{R}$
- 后空翻：根高度 h_t 和根俯仰角 $\theta_{\text{pitch}t} \in \mathbb{R}$
- 侧空翻：根高度 h_t 和根横滚角 $\theta_{\text{roll}t} \in \mathbb{R}$

D. 辅助力设计

为每个任务设计了辅助力模式。施加时机在所有任务中共享，为 $T = \{(1.0\text{秒至}1.1\text{秒})\}$ ，而施加点 P 和力向量 F 则因任务而异。辅助力模式如图4所示。

a) 跳跃:

$$\begin{aligned} P^{\text{jump}} &= \{\mathbf{p}_{\text{FL}}^{\text{Scapula}}, \mathbf{p}_{\text{FR}}^{\text{Scapula}}, \mathbf{p}_{\text{BL}}^{\text{Scapula}}, \mathbf{p}_{\text{BR}}^{\text{Scapula}}\}, \\ F^{\text{jump}} &= \{(0, 0, f_{\text{jump}}(h^{\text{target}})/4)\} \end{aligned}$$

此处， $f_{\text{jump}}(h^{\text{target}})$ 是由目标高度 h^{target} 决定的辅助力大小。该值通过将跳跃运动建模为简单抛体运动，并计算达到目标高度所需初速度所需的平均辅助力而得。

b) 后空翻:

$$\begin{aligned} P^{\text{backflip}} &= \{\mathbf{p}_{\text{FL}}^{\text{Scapula}}, \mathbf{p}_{\text{FR}}^{\text{Scapula}}\}, \\ F^{\text{backflip}} &= \{(0, 0, 175\text{N})\} \end{aligned}$$

c) 侧空翻:

$$\begin{aligned} P^{\text{lateral}} &= \{\mathbf{p}_{\text{FR}}^{\text{Scapula}}, \mathbf{p}_{\text{BR}}^{\text{Scapula}}\}, \\ F^{\text{lateral}} &= \{(0, 0, 300\text{N})\} \end{aligned}$$

E. 自适应课程设计

采用基于成功率的课程调度策略来衰减辅助力。算法1中的成功率根据以下标准计算。

a) 跳跃:

$$|h_t^{\text{max}} - h^{\text{target}}| < 0.1 \quad \wedge \quad |h_t| < 0.1$$

b) 后空翻、侧空翻:

$$|\theta_t - 2\pi| < 0.3 \quad \wedge \quad |h_t| < 0.1$$

所有任务的成功率阈值设置为 $\zeta = 0.6$ ，衰减步长设置为 $\varepsilon = 0.01$ 。

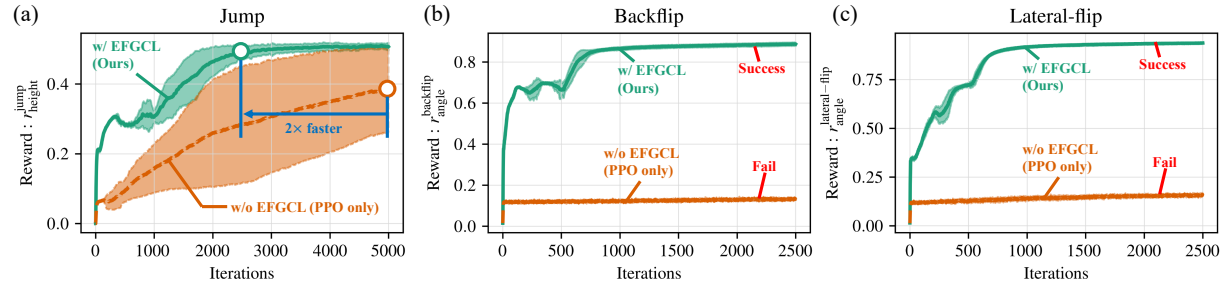


Fig. 5. Comparison of learning curves with and without EFGCL. (a) Jump, (b) Backflip, (c) Lateral-flip. With EFGCL, learning progresses stably and converges to high reward values in all tasks. Without EFGCL, Jump exhibits large reward variance, while learning hardly progresses for Backflip and Lateral-flip.

F. Teacher-Student Learning

Markovianity is a crucial property in reinforcement learning environments. Following prior work [2], we adopt a Teacher-Student learning architecture.

Teacher-Student learning consists of two stages: reinforcement learning of the teacher policy and supervised learning of the student policy, where the teacher policy acts as a supervisor. During supervised learning, the student policy is trained to minimize an action-matching loss and a reconstruction loss on privileged observations. Since the proposed method is integrated with reinforcement learning, EFGCL is applied only during the training of the teacher policy. The overall learning structure is illustrated in Fig. 2.

V. EXPERIMENT AND RESULT

A. Learning Performance and Real Robot Deployment

We compared the proposed EFGCL with a PPO baseline over 10 random seeds. As shown in Fig. 5, EFGCL achieved stable convergence and high rewards across all tasks (Jump, Backflip, and Lateral-flip). In contrast, the baseline failed to learn the flipping tasks and exhibited high variance in the Jump task. As visualized in Fig. 6, the baseline often resulted in unnatural postures, whereas EFGCL acquired natural dynamic motions. This stable learning process was facilitated by the adaptive curriculum, which automatically adjusted the assistive force decay based on the success rate (Fig. 7). Furthermore, the policies learned via EFGCL were distilled and deployed on the quadrupedal robot KLEIYN. As shown in Fig. 8, the dynamic motions observed in simulation were successfully reproduced on the real robot for all three tasks.

B. Ablation Study of EFGCL Force Design

To evaluate the sensitivity of EFGCL to heuristic design choices, we varied the application point, magnitude, and timing of the assistive force in the Backflip task. The results in Fig. 9 demonstrate that learning is robust over a wide range of parameters. Successful policies were acquired even when the force was applied to different links (thigh or calf) or when the magnitude varied within a reasonable range (140–210 N). Learning failed only in extreme cases where the assistance was physically insufficient (e.g., 100 N) or excessive (e.g., 250 N), or when the application timing was too short (1.0 s



Fig. 6. Snapshots of learned motions with and without EFGCL. (a–c) With EFGCL, (d–f) without EFGCL. With EFGCL, natural and stable motions are acquired, whereas without EFGCL, unnatural postures and failure to achieve the target motions are observed.

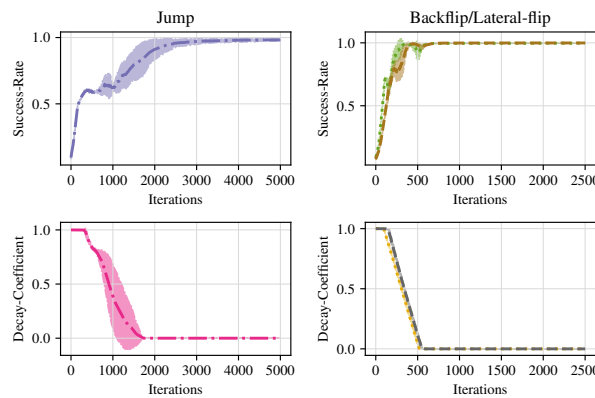


Fig. 7. Transitions of the success rate and assistive force decay factor in EFGCL. (a) Jump task, (b) Backflip and Lateral-flip tasks. The decay speed of the assistive force is automatically adjusted according to the success rate.

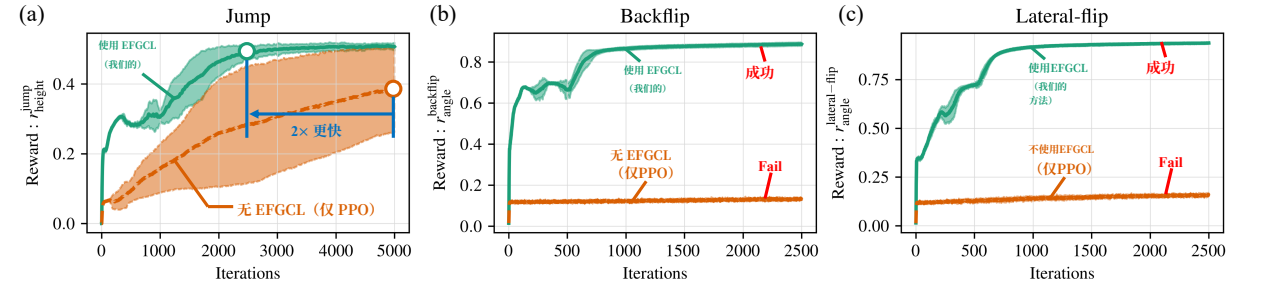


图5. 有无EFGCL的学习曲线对比。(a) 跳跃, (b) 后空翻, (c) 侧空翻。使用EFGCL时, 学习过程稳定推进, 并在所有任务中收敛到高奖励值。未使用EFGCL时, 跳跃表现出较大的奖励方差, 而后空翻和侧空翻的学习几乎无法推进。

F. 师生学习

马尔可夫性是强化学习环境中的一个关键属性。遵循先前的工作 [2], 我们采用了师生学习架构。

师生学习包括两个阶段: 教师策略的强化学习和学生策略的监督学习, 其中教师策略充当监督者。在监督学习阶段, 学生策略通过最小化动作匹配损失和特权观测上的重建损失进行训练。由于所提方法与强化学习相结合, EFGCL仅在教师策略训练期间应用。整体学习结构如图2所示。

V. 实验与结果

A. 学习性能与真实机器人部署

我们将所提出的EFGCL与PPO基线在10个随机种子上进行了比较。如图5所示, EFGCL在所有任务(跳跃、后空翻和侧空翻)中均实现了稳定收敛和高奖励。相比之下, 基线在翻转任务中未能学习成功, 并且在跳跃任务中表现出高方差。如图6所示, 基线常常导致不自然的姿态, 而EFGCL则习得了自然的动态运动。这一稳定的学习过程得益于自适应课程, 它根据成功率自动调整辅助力衰减(图7)。此外, 通过EFGCL学习到的策略被蒸馏并部署到四足机器人KLEIYN上。如图8所示, 仿真中观察到的动态运动在真实机器人上成功复现了所有三项任务。

B. EFGCL力设计的消融研究

为了评估EFGCL对启发式设计选择的敏感性, 我们在后空翻任务中改变了辅助力的施加点、大小和时机。图9中的结果表明, 学习在广泛的参数范围内具有鲁棒性。即使将力施加到不同的连杆(大腿或小腿)上, 或者大小在合理范围内(140–210 N)变化, 也能获得成功的策略。仅在辅助力极端情况下学习才会失败

物理上不足(例如100 N)或过大(例如250 N), 或者施加时机过短(1.0秒

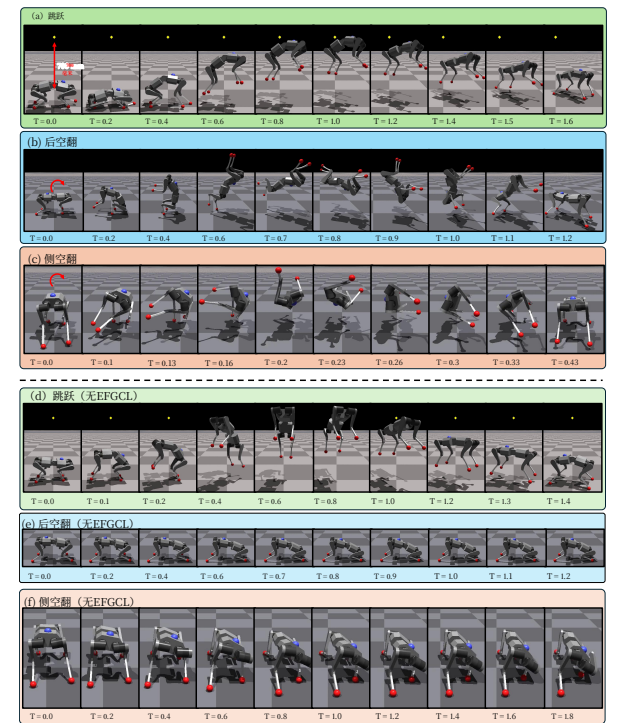


图6. 使用和不使用EFGCL学习到的运动快照。(a–c) 使用EFGCL, (d–f) 不使用EFGCL。使用EFGCL时, 能够获得自然且稳定的运动; 而不使用EFGCL时, 则观察到不自然的姿态以及未能实现目标运动的情况。

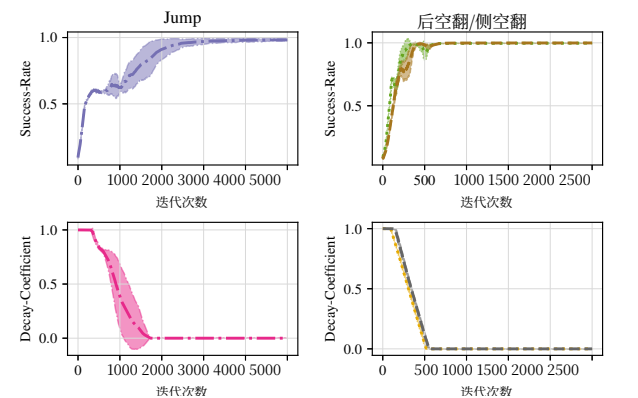


图7. EFGCL中成功率和辅助力衰减因子的变化。(a) 跳跃任务, (b) 后空翻和侧空翻任务。辅助力的衰减速度根据成功率自动调整。



Fig. 8. Reproduction of learned motions on the real quadrupedal robot. (a) Jump, (b) Backflip, (c) Lateral-flip.

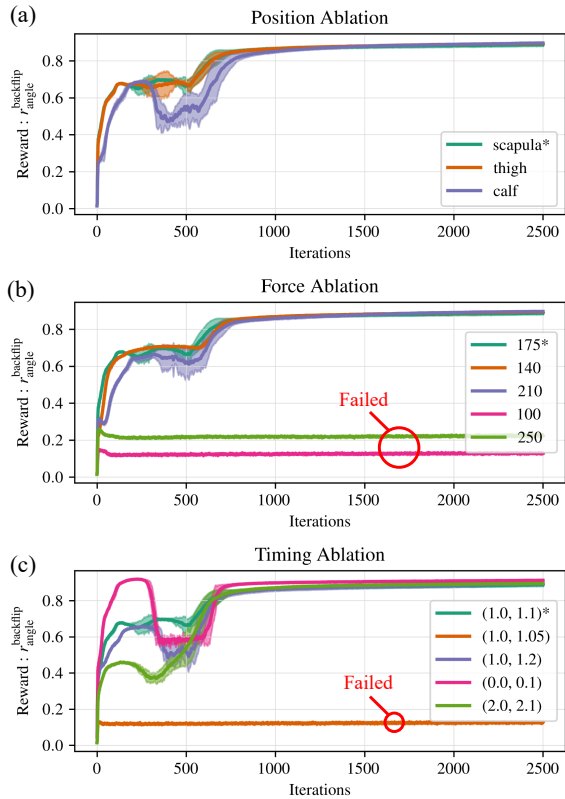


Fig. 9. Ablation study on assistive force design. (a) Application points, (b) force magnitudes, (c) application timing. The asterisks (*) in the legends indicate the conditions used in the main experiments.

to 1.05 s). These results indicate that precise tuning is not required, as long as the assistance roughly facilitates the target motion.

C. Evaluation of Accelerated Critic Value Estimation

To validate the hypothesis that external guidance accelerates critic learning, we analyzed value estimates during the Jump task. Fig. 10 compares the value function outputs for a successful reference motion at different training stages. With EFGCL, the value estimates converged to the final distribution as early as 200 iterations. In contrast, the baseline required more than 1,000 iterations to reach a comparable

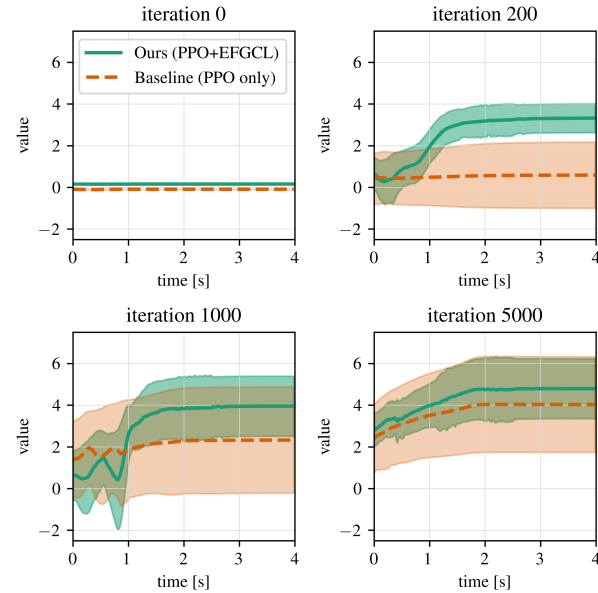


Fig. 10. Comparison of value function estimation with and without EFGCL. With EFGCL, value estimates quickly converge to distributions close to the final value function, demonstrating accelerated and stabilized value estimation.

level of accuracy and exhibited larger variance. This result confirms that experiencing successful states early in training significantly accelerates value function estimation.

VI. DISCUSSION

A. Efficacy and Robustness of Guided Exploration

The experimental results demonstrate that EFGCL significantly stabilizes the learning of dynamic motion skills by accelerating value function estimation during the early training phase. Unlike reward shaping or imitation learning, which rely on complex reward design or expert datasets, EFGCL guides exploration through direct physical assistance in the form of external forces.

Furthermore, the ablation study shows that the proposed method is highly robust to variations in assistive force design. As long as the assistance roughly facilitates the target motion, learning can succeed without precise parameter tuning. These results suggest that the principle of “physically experiencing success” provides a general and cost-effective strategy for overcoming exploration challenges in dynamic robotic reinforcement learning.

B. Limitations and Future Work

This study focuses on validating the principle of artificially enabling agents to experience successful motions. Accordingly, the assistive forces were designed based on task-specific physical intuition. While such heuristic designs are sufficient for the single-shot dynamic motion tasks considered in this work, the design burden may increase for more complex and continuous motions.

For continuous behaviors such as dancing, maintaining the overall motion structure often requires learning based on reference trajectories. In such cases, using the proposed framework of external forces with sparse rewards alone may be insufficient, and combining it with trajectory-tracking reward designs or imitation learning is likely to be more



图8. 在真实四足机器人上复现学习到的运动。(a) 跳跃, (b) 后空翻, (c) 侧空翻。

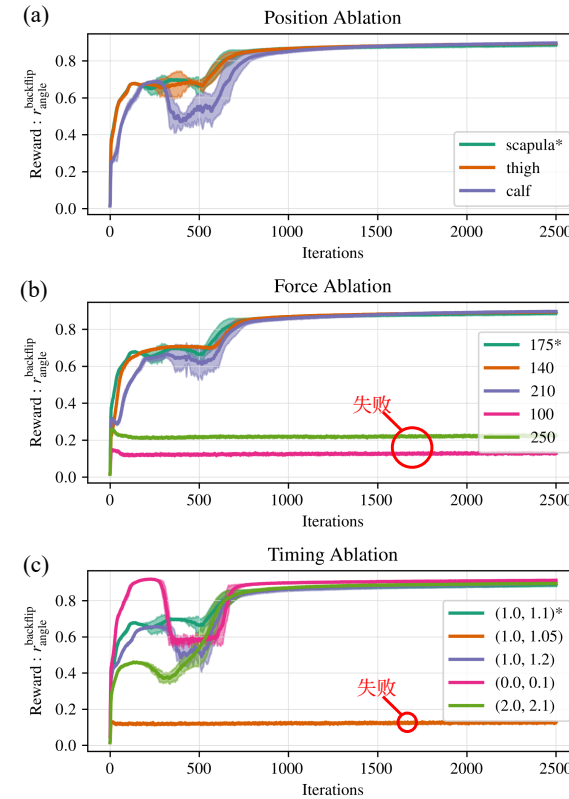


图9. 辅助力设计的消融研究。(a) 施加点, (b) 力的大小, (c) 施加时机。图例中的星号 (*) 表示主要实验中使用的条件。

至1.05秒)。这些结果表明，只要辅助大致促进目标运动，就无需精确调参。

C. 加速评论家价值估计的评估

为验证外部引导能加速评论家学习的假设，我们分析了跳跃任务中的价值估计。图10比较了不同训练阶段下，一个成功参考运动的价值函数输出。使用EFGCL时，价值估计早在200次迭代时就收敛到了最终分布。相比之下，基线方法需要超过1000次迭代才能达到相当的

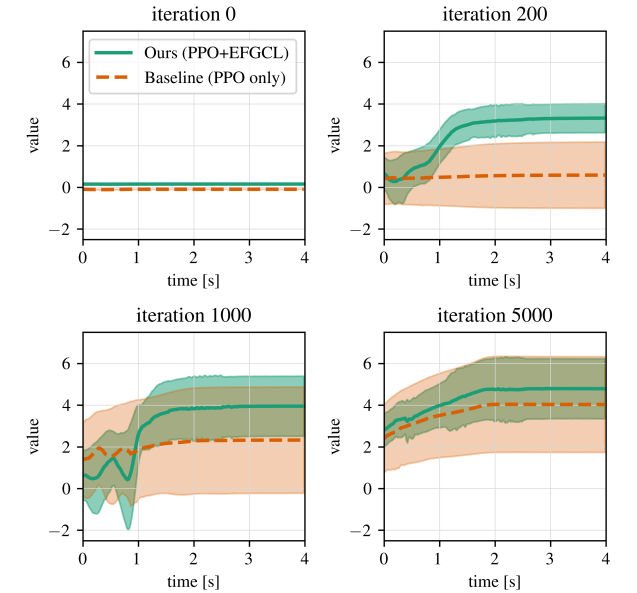


图10. 使用和不使用EFGCL的价值函数估计对比。使用EFGCL时，价值估计迅速收敛到接近最终价值函数的分布，展示了加速且稳定的价值估计。

准确度水平，且表现出更大的方差。这一结果证实，在训练早期体验成功状态能显著加速价值函数估计。

VI. 讨论

A. 引导式探索的有效性与鲁棒性

实验结果表明，EFGCL通过在早期训练阶段加速价值函数估计，显著稳定了动态运动技能的学习。与依赖复杂奖励设计或专家数据集的奖励塑形或模仿学习不同，EFGCL通过外力形式的直接物理辅助来引导探索。

此外，消融研究表明，所提出的方法对辅助力设计的变化具有高度鲁棒性。只要辅助力大致促进目标运动，无需精确的参数调整即可成功学习。这些结果表明，“物理体验成功”的原则为克服动态机器人强化学习中的探索挑战提供了一种通用且经济有效的策略。

B. 局限性与未来工作

本研究聚焦于验证人为使智能体体验成功运动这一原理的有效性。因此，辅助力是基于任务特定的物理直觉设计的。尽管此类启发式设计足以应对本工作中所考虑的单次动态运动任务，但对于更复杂和连续的运动，设计负担可能会增加。

对于舞蹈等连续行为，保持整体运动结构通常需要基于参考轨迹的学习。在这种情况下，仅使用所提出的外力框架配合稀疏奖励可能不够充分，将其与轨迹跟踪奖励设计或模仿学习相结合可能会更加

effective. Developing automatic optimization or generation methods for assistive forces that remain effective for complex target motions is an important direction for future work.

VII. CONCLUSION

Inspired by spotting in gymnastics, we proposed External Force Guided Curriculum Learning (EFGCL), a reinforcement learning framework that guides exploration through decaying external forces. Without relying on complex reward shaping or reference trajectories, EFGCL enables a quadrupedal robot to acquire dynamic whole-body motions, such as jumping and flipping, that are difficult for standard RL methods.

Through both simulation and real-robot experiments, we demonstrated successful sim-to-real transfer and showed that physical assistance accelerates value function estimation by allowing the agent to experience successful states early in training. Although the current approach relies on heuristic force design, the results suggest that *physical guidance* represents a promising and general paradigm for guided exploration, complementary to reward-based and imitation-based methods, in the learning of complex whole-body motions.

APPENDIX I REWARD DEFINITIONS

Table I summarizes the reward terms shared across all tasks and their definitions. Here, P_{col} denotes the set of link indices used to detect collisions with the ground. This set includes 14 links in total: the body, scapula, thigh, and calf, excluding the feet. The indicator function δ_t^{term} takes the value of 1 if the episode terminates due to the trunk contacting the ground at time t , and 0 otherwise.

A. Task-Specific Target Variables

Table II presents the task-specific definitions of the target variable x_t , the target value x^{target} , and the normalization coefficient s_x used in the task progress reward ρ_t^{task} shown in Table I.

TABLE I SUMMARY OF REWARD TERMS SHARED ACROSS ALL TASKS.	
Reward term	Definition
Task progress	$\rho_t^{\text{task}} = \exp(-\ x_t - x^{\text{target}}\ ^2 / s_x)$
Standing	$\rho_t^{\text{stand}} = \exp\left(-\frac{\ h_t\ ^2}{0.01}\right) + \exp\left(-\frac{\ \mathbf{q}_t - \mathbf{q}^{\text{stand}}\ ^2}{0.25}\right)$
Angular regularization	$r_t^{\text{ang}} = -\ (\omega_t^{\text{non-target}})^2\ $
Collision penalty	$-1.0 \times \sum_{i \in P_{col}} (f_{i,z} > 0.1)$
Termination penalty	$-100 \times \delta_t^{\text{term}}$
Joint velocity penalty	$-5 \times 10^{-4} \ \dot{\mathbf{q}}_t\ ^2$
Joint acceleration penalty	$-1 \times 10^{-7} \ \ddot{\mathbf{q}}_t\ ^2$

REFERENCES

- [1] J. Hwangbo, J. Lee, A. Dosovitskiy, D. Bellicoso, V. Tsounis, V. Koltun, and M. Hutter, “Learning agile and dynamic motor skills for legged robots,” *Science Robotics*, vol. 4, no. 26, 2019.
- [2] T. Miki, J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, “Learning robust perceptive locomotion for quadrupedal robots in the wild,” *Science Robotics*, vol. 7, no. 62, 2022.

TABLE II
TASK-SPECIFIC INSTANTIATIONS OF THE TASK PROGRESS REWARD.

Task	Target variable x_t	x^{target}	s_x
Jump	h_t^{max}	h^{target}	0.01
Backflip	θ_t^{pitch}	2π	π^2
Lateral-Flip	θ_t^{roll}	2π	π^2

- [3] Z. Zhuang, Z. Fu, J. Wang, C. Atkeson, S. Schwertfeger, C. Finn, and H. Zhao, “Robot parkour learning,” in *Conference on Robot Learning (CoRL)*, 2023.
- [4] D. Vogel, R. Baines, J. Church, J. Lotzer, K. Werner, and M. Hutter, “Robust ladder climbing with a quadrupedal robot,” in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 7239–7244.
- [5] H. Kim, H. Oh, J. Park, Y. Kim, D. Youm, M. Jung, M. Lee, and J. Hwangbo, “High-speed control and navigation for quadrupedal robots on complex and discrete terrain,” *Science Robotics*, vol. 10, no. 102, p. eads6192, 2025.
- [6] K. Yoneda, K. Kawaharazuka, T. Suzuki, T. Hattori, and K. Okada, “Kleyn : A quadruped robot with an active waist for both locomotion and wall climbing,” in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 8783–8789.
- [7] J. Eßer, N. Bach, C. Jestel, O. Urbann, and S. Kerner, “Guided reinforcement learning: A review and evaluation for efficient and effective real-world robotics [survey],” *IEEE Robotics & Automation Magazine*, vol. 30, no. 2, pp. 67–85, 2022.
- [8] X. B. Peng, P. Abbeel, S. Levine, and M. Van de Panne, “Deepmimic: Example-guided deep reinforcement learning of physics-based character skills,” *ACM Transactions On Graphics (TOG)*, vol. 37, no. 4, pp. 1–14, 2018.
- [9] X. B. Peng, E. Coumans, T. Zhang, T.-W. E. Lee, J. Tan, and S. Levine, “Learning agile robotic locomotion skills by imitating animals,” in *Robotics: Science and Systems*, 07 2020.
- [10] J. Wu, G. Xin, C. Qi, and Y. Xue, “Learning robust and agile legged locomotion using adversarial motion priors,” *IEEE Robotics and Automation Letters*, vol. 8, no. 8, pp. 4975–4982, 2023.
- [11] Y. Fuchioka, Z. Xie, and M. Van de Panne, “Opt-mimic: Imitation of optimized trajectories for dynamic quadruped behaviors,” in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 5092–5098.
- [12] C. Li, M. Vlastelica, S. Blaes, J. Frey, F. Grimmeringer, and G. Martius, “Learning agile skills via adversarial imitation of rough partial demonstrations,” in *Conference on Robot Learning*. PMLR, 2023, pp. 342–352.
- [13] S. Devlin and D. Kudenko, “Theoretical considerations of potential-based reward shaping for multi-agent systems,” in *Tenth international conference on autonomous agents and multi-agent systems*. ACM, 2011, pp. 225–232.
- [14] V. Atanassov, J. Ding, J. Kober, I. Havoutis, and C. Della Santina, “Curriculum-based reinforcement learning for quadrupedal jumping: A reference-free design,” *IEEE Robotics & Automation Magazine*, 2024.
- [15] G. Bellegarda, C. Nguyen, and Q. Nguyen, “Robust quadruped jumping via deep reinforcement learning,” *Robotics and Autonomous Systems*, vol. 182, p. 104799, 2024.
- [16] G. Vasan, Y. Wang, F. Shahriar, J. Bergstra, M. Jägersand, and A. R. Mahmood, “Revisiting sparse rewards for goal-reaching reinforcement learning,” *Reinforcement Learning Journal*, vol. 4, pp. 1841–1854, 2024.
- [17] S. Sorzano, “How spotting with touch affects skill performance and self confidence in gymnasts,” Master’s thesis, Trent University (Canada), 2023.
- [18] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” 2017.
- [19] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, “High-dimensional continuous control using generalized advantage estimation,” in *International Conference on Learning Representations (ICLR)*, 2016.
- [20] J. Liang, V. Makoviychuk, A. Handa, N. Chentanez, M. Macklin, and D. Fox, “Gpu-accelerated robotic simulation for distributed reinforcement learning,” in *Conference on Robot Learning (CoRL)*, 2018.

有效。开发能够对复杂目标运动保持有效的辅助力自动优化或生成方法，是未来工作的重要方向。

七、结论

受体操运动中保护的启发，我们提出了外力引导课程学习（EFGCL），这是一种通过衰减外力引导探索的强化学习框架。在不依赖复杂奖励塑形或参考轨迹的情况下，EFGCL使四足机器人能够获得标准强化学习方法难以实现的动态全身运动，如跳跃和翻转。

通过仿真和真实机器人实验，我们证明了成功的仿真到现实迁移，并表明物理辅助通过让智能体在训练早期体验成功状态来加速价值函数估计。尽管当前方法依赖于启发式力设计，但结果表明，物理引导代表了一种有前景且通用的引导式探索范式，与基于奖励和基于模仿的方法互补，适用于复杂全身运动的学习。

附录一 奖励定义

表一总结了所有任务共用的奖励项及其定义。此处， P_{col} 表示用于检测与地面碰撞的连杆索引集合。该集合共包含14个连杆：身体、肩胛骨、大腿和小腿，不包括脚。指示函数 δ_t^{term} 在回合因躯干在时间 t 接触地面而终止时取值为1，否则为0。

A. 任务特定目标变量

表II展示了任务进度奖励 ρ_t^{task} 中使用的目标变量 x_t 、目标值 x^{target} 和归一化系数 s_x 的任务特定定义，如表I所示。

表I 所有任务共用的奖励项汇总	
奖励项	定义
任务进度	$\rho_t^{\text{task}} \equiv \frac{\exp\left(-\frac{\ x_t - x^{\text{target}}\ ^2}{0.01}\right)}{\exp\left(-\frac{\ x_t - x^{\text{target}}\ ^2}{0.01}\right) + \exp\left(-\frac{\ x_t - x^{\text{target}}\ ^2}{0.25}\right)}$
站立	$r_t^{\text{ang}} = -\ (\omega_t^{\text{non-target}})^2\ $
角速度正则化	$-1.0 \times \sum_{i \in P_{col}} (f_{i,z} > 0.1)$
碰撞惩罚	$-100 \times \delta_t^{\text{term}}$
终止惩罚	$-5 \times 10^{-4} \ \dot{\mathbf{q}}_t\ ^2$
关节速度惩罚	$-1 \times 10^{-7} \ \ddot{\mathbf{q}}_t\ ^2$
关节加速度惩罚	

参考文献

- [1] J. Hwangbo, J. Lee, A. Dosovitskiy, D. Bellicoso, V. Tsounis, V. Koltun和M. Hutter, “学习腿式机器人的敏捷和动态运动技能”，*Science Robotics*, 第4卷，第26期，2019年。[2] T. Miki, J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun和M. Hutter, “在野外学习四足机器人的鲁棒感知运动”，*Science Robotics*, 第7卷，第62期，2022年。

表II 任务进度奖励的任务特定实例化。			
Task	目标变量 x_t	x^{target}	s_x
Jump	h_t^{max}	h^{target}	0.01
后空翻	θ_t^{pitch}	2π	π^2
侧空翻	θ_t^{roll}	2π	π^2

- [3] Z. Zhuang, Z. Fu, J. Wang, C. Atkeson, S. Schwertfeger, C. Finn和H. Zhao, “机器人跑酷学习”，载于机器人学习会议（*CoRL*），2023年。[4] D. Vogel、R. Baines、J. Church、J. Lotzer、K. Werner和M. Hutter, “稳健爬梯：四足机器人的应用”，载于2025年*IEEE/RSJ*智能机器人与系统国际会议（*IROS*），2025年，第7239–7244页。[5] H. Kim、H. Oh、J. Park、Y. Kim、D. Youm、M. Jung、M. Lee和J. Hwangbo, “复杂离散地形上四足机器人的高速控制与导航”，*Science Robotics*, 第10卷，第102期，第eads6192页，2025年。[6] K. Yoneda、K. Kawaharazuka、T. Suzuki、T. Hattori和K. Okada, “Kleyn：一款兼具运动与爬墙能力的主动腰部四足机器人”，载于2025年*IEEE/RSJ*智能机器人与系统国际会议（*IROS*），2025年，第8783–8789页。[7] J. Eßer、N. Bach、C. Jestel、O. Urbann和S. Kerner, “引导强化学习：面向高效实用现实世界机器人学的综述与评估 [调查],”，《*IEEE*机器人与自动化杂志》，第30卷，第2期，第67–85页，2022年。[8] X. B. Peng、P. Abbeel、S. Levine和M. Van de Panne, “Deepmimic: 基于物理的角色技能的示例引导深度强化学习”，《*ACM*图形学汇刊》（*TOG*），第37卷，第4期，第1–14页，2018年。[9] X. B. Peng、E. Coumans、T. Zhang、T.-W. E. Lee、J. Tan和S. Levine, “通过模仿动物学习敏捷机器人运动技能”，载于机器人学：科学与系统，2020年7月。[10] J. Wu、G. Xin、C. Qi和Y. Xue, “利用对抗性运动先验学习稳健敏捷的腿式运动”，《*IEEE*机器人与自动化快报》，第8卷，第8期，第4975–4982页，2023年。[11] Y. Fuchioka、Z. Xie和M. Van de Panne, “Opt-mimic: 动态四足行为的优化轨迹模仿”，载于2023年*IEEE*国际机器人与自动化会议（*ICRA*），2023年，第5092–5098页。[12] C. Li、M. Vlastelica、S. Blaes、J. Frey、F. Grimmeringer和G. Martius, “通过对抗性模仿粗糙的部分演示学习敏捷技能”，载于机器人学习会议。PMLR, 2023年，第342–352页。[13] S. Devlin和D. Kudenko, “多智能体系统中基于势能的奖励塑形的理论考量”，载于第十届国际自主智能体与多智能体系统会议。ACM, 2011年，第225–232页。[14] V. Atanassov、J. Ding、J. Kober、I. Havoutis和C. Della Santina, “基于课程表的强化学习用于四足跳跃：一种无参考设计”，《*IEEE*机器人与自动化杂志》，2024年。[15] G. Bellegarda、C. Nguyen和Q. Nguyen, “通过深度强化学习实现稳健四足跳跃”，《机器人与自主系统》，第182卷，第104799页，2024年。[16] G. Vasan、Y. Wang、F. Shahriar、J. Bergstra、M. Jägersand和A. R. Mahmood, “重新审视目标达成强化学习中的稀疏奖励”，《强化学习杂志》，第4卷，第1854页，2024年，第1841–1854页。[17] S. Sorzano, “触摸保护如何影响体操运动员的技能表现和自信心”，硕士论文，特伦特大学（加拿大），2023年。

[18] J. Schulman, F. Wolski, P. Dhariwal, A. Radford和O. Klimov, “近端策略优化算法”，2017年。[19] J. Schulman, P. Moritz, S. Levine, M. Jordan和P. Abbeel, “使用广义优势估计的高维连续控制”，载于国际学习表征会议（*ICLR*），2016年。[20] J. Liang, V. Makoviychuk, A. Handa, N. Chentanez, M. Macklin和D. Fox, “用于分布式强化学习的GPU加速机器人仿真”，载于机器人学习会议（*CoRL*），2018年。